St. Vincent Pallotti
College of Engineering & Technology
(An Autonomous Institute)

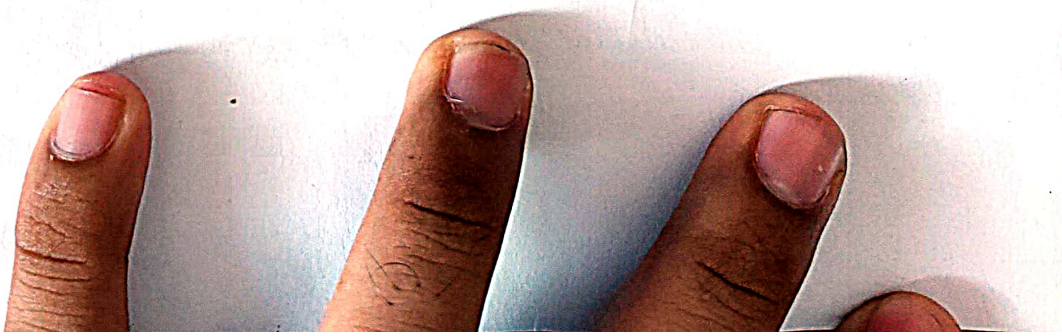
Academic Year: 2024-25

Mid Sem Examination (MSE) - B. Tech

Course: -	**Business strategies**	**Course Code: -**	22CB504T
		Max. Marks: -	20
Semester: -	**V**	**Duration: -**	1hr
Program: -	**Computer Science and Business Systems**	**Date of Paper**	06/09/2024

Instructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) All Questions carry marks as indicated
5) Use of non-programmable calculator is allowed.

Oue. No.	Description of Question	Marks	[CO]	[BTL]
Que.1(a)	Define Strategic Management.	02	01	01
Que.1(b)	Write short note on 'fit concept' in Strategic Management? Give one example.	02	01	01
Que.1(c)	What do you understand by Core competencies?	01	02	01
Que.2(a)	Explain in detail the objectives of Strategic Management.	04	01	02
Que.2(b)	Write down sources of Sustained Competitive Advantage. Explain any 5 in detail.	04	01	02
	OR			
Que.3(a)	Discuss in detail – Schools of thought in Strategic Management.	04	01	02
Que.3(b)	Explain in detail how Core Competencies can be the roots for Competitive Advantage?	04	02	02
Que.4(a)	Write down in detail the McKinsey's 7 S framework.	04	02	02
Que.4(b)	Discuss the steps involved in content strategy with examples	03	02	02
	OR			
Que.5(a)	Differenciate between Competitive Advantage and Core Competence Tabular Form	04	02	02
Que.5(b)	Elaborate the statement – 'Strategic Management is the amalgamation of two words: Strategy and Management'.	03	02	02



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[BTL]
1

Course: -
Semester: -
Progr-
Academic Year: 2024-2-
Mid Sem Examinatio-
Behavioral S-

St. Vincent Pallotti
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Academic Year: 2024-25
Mid Sem Examination (MSE) - B. Tech

Course: -	**Elective-I Cloud, Micro services & Applications**	**Course Code: -**	22CB506T
Semester: -	**V**	**Max. Marks: -**	35
Program: -	**Computer Science & Business Systems**	**Duration: -**	1 Hour 30 Minutes
		Date of Paper	10/09/2024

Instructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) All Questions carry marks as indicated
6) Use of non-programmable calculator is allowed.

Que. No.	Description of Question	Marks	[CO]	[BTL]
Que.1(a)	Define Cloud Computing.	02	1	1
Que.1(b)	What is Virtualization?	02	2	1
Que.1(c)	Enlist any 3 Characteristics of Cloud Computing.	02	1	1
Que.1(d)	What are the types of Service Level Agreement?	02	2	1
Que.1(e)	Outline the benefits of Identity and Access Management.	02	3	2
Que.2(a)	Depict and describe Cloud Computing Architecture.	05	1	2
Que.2(b)	Discuss how Cloud Computing is used in Education with the help of suitable diagram.	05	1	3
	OR			
Que.3(a)	Illustrate the NIST Model.	06	1	2
Que.3(b)	Identify advantages and disadvantages of Cloud Computing?	04	1	3
Que.4(a)	Outline the Purpose of Hypervisor. Explain the types of Hypervisor with the help of diagram.	06	2	2
Que.4(b)	Represent overall MapReduce word count process for the input given below. Fuzzy Wuzzy was a bear. Fuzzy Wuzzy had no hair. Fuzzy Wuzzy wasn't very fuzzy, was he?	04	2	3
	OR			
Que.5(a)	Describe any two algorithms used for load balancing with the help of diagrams.	05	2	2
Que.5(b)	Plot HDFS Structure with all the components and describe role of each component.	05	2	1

Que.6	Describe IAM components. Enlist features of IAM.	05	3	2
	OR			
Que.7	Discuss the importance of various components of Service Level Agreement.	05	3	2

Course Outcomes -

CO1	Understand Cloud, various cloud service and deployment models, enabling students to choose appropriate cloud solutions based on organizational needs.
CO2	Explain the role and significance of virtualization technologies and how they integrate with cloud computing to enhance resource management and scalability.
CO3	Identify and assess cloud security risks, and implement fundamental security measures to protect cloud environments.
CO4	Design and develop microservices-based architectures that adhere to cloud-native principles, improving application scalability and maintainability.
CO5	Apply DevOps tools and practices to streamline cloud application development and deployment processes.



Academic Year: 2024-25
Mid Sem Examination (MSE)
S OF MANAGEMENT
Max. Marks:
Duration: -
Date of Paper 05/09/24
[BTL]

St. Vincent Pallotti College of Engineering & Technology

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Academic Year: 2024-25
Mid Sem Examination (MSE) - B. Tech

Course: -	**Behavioral Economics**	**Course Code: -**	22CB507T
Semester: -	**V**	**Max. Marks: -**	35
Program: -	**Computer Science & Business Systems**	**Duration: -**	1hr 30min
		Date of Paper	12/09/24

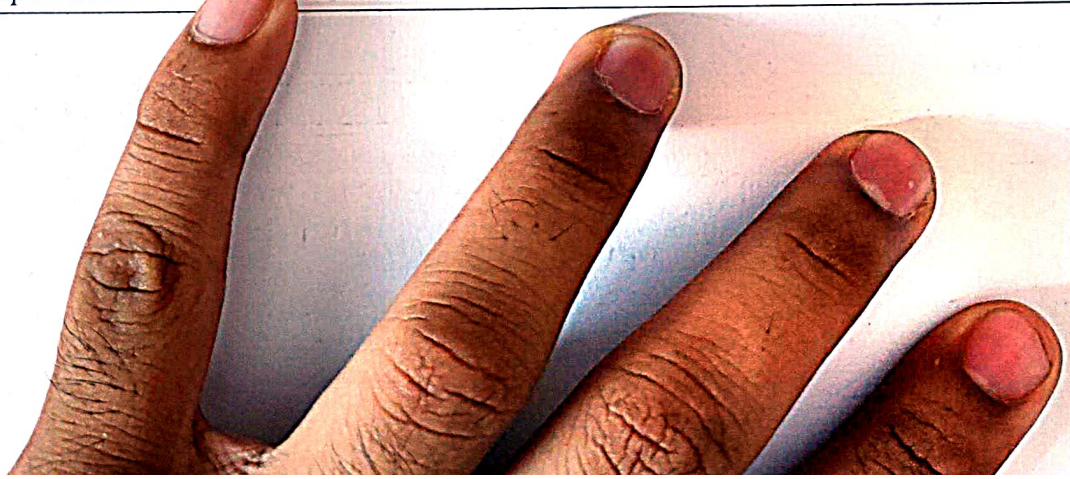
Instructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) All Questions carry marks as indicated
6) Use of non-programmable calculator is allowed.

Que. No.	Description of Question	Marks	[CO]	[BTL]
Que.1(a)	Select at least one option for the statement below. The term "homo economicus" is used to describe: 1. Infinite ability to make rational decisions 2. Irrational decision making 3. Infinite ability to make irrational decisions 4. Finite ability to make rational decisions	02	1	1
Que.1(b)	Select at least one option for the statement below. If a consumer buys a lottery ticket but also purchases fire insurance, then he or she can be said to be: 1. Risk loving in losses, risk averse in gains 2. Risk neutral in losses and gains 3. Risk averse in losses, risk loving in gains 4. Risk loving in losses, risk averse in gains	02	1	2
Que.1(c)	Select at least one option for the statement below. The government sent a text message to tax payers with the reminder that "most people pay on time to avoid penalties". This type of message is taking account of what concept in behavioral economics? 1. Risk loving 2. Anchoring 3. Bounded rationality 4. Nudging	02	2	2
Que.1(d)	Select at least one option for the statement below. A person is said to be bounded rational if: 1. She will always make the best decision, independent of the number of option available 2. She is able to only assess some of the possibilities that could occur in any given circumstance 3. Given the choice between 6 options or 12 options, 12 is always better 4. She is able to asses all the bounds of possibilities that could occur in any given scenario.	02	2	2

Que.1(c)	Define rational behavior in economics?	02	3	2
Que.2(a)	Explain Prospect Theory and illustrate its principles with examples.	05	1	2
Que.2(b)	Compare and contrast behavioral economics with classical economics.	05	1	4
	OR			
Que.3(a)	Discuss the phenomenon of Money Illusion and offer examples that demonstrate its significance.	05	1	2
Que.3(b)	Relate Charitable donation, Altruism and behavioral economics.	05	1	4
Que.4(a)	Identify and discuss the core axioms of Choice Theory.	05	2	2
Que.4(b)	Examine the different models of decision making and explain their main attributes.	05	2	2
	OR			
Que.5(a)	How does the concept of retail therapy relate to choice theory? Explain its connection.	05	2	2
Que.5(b)	Describe Public Choice Theory and provide examples to demonstrate its key concepts.	05	2	2
Que.6	What is Anchoring Effect? Explain with example.	05	3	2
	OR			
Que.7	Provide an explanation of heuristics in decision-making and demonstrate their application with an example.	05	3	2

Course Outcomes -

CO1	Discuss the basic concepts of Economic theory.
CO2	Students will be able to understand various notions in traditional Microeconomics.
CO3	Apply the concepts of traditional and modern economics like beliefs, biases and choice while focusing on decision making.
CO4	Develop holistic understanding of choice in economics.
CO5	Compare and interco[illegible]raditional and modern microeconomics with strategic choice.



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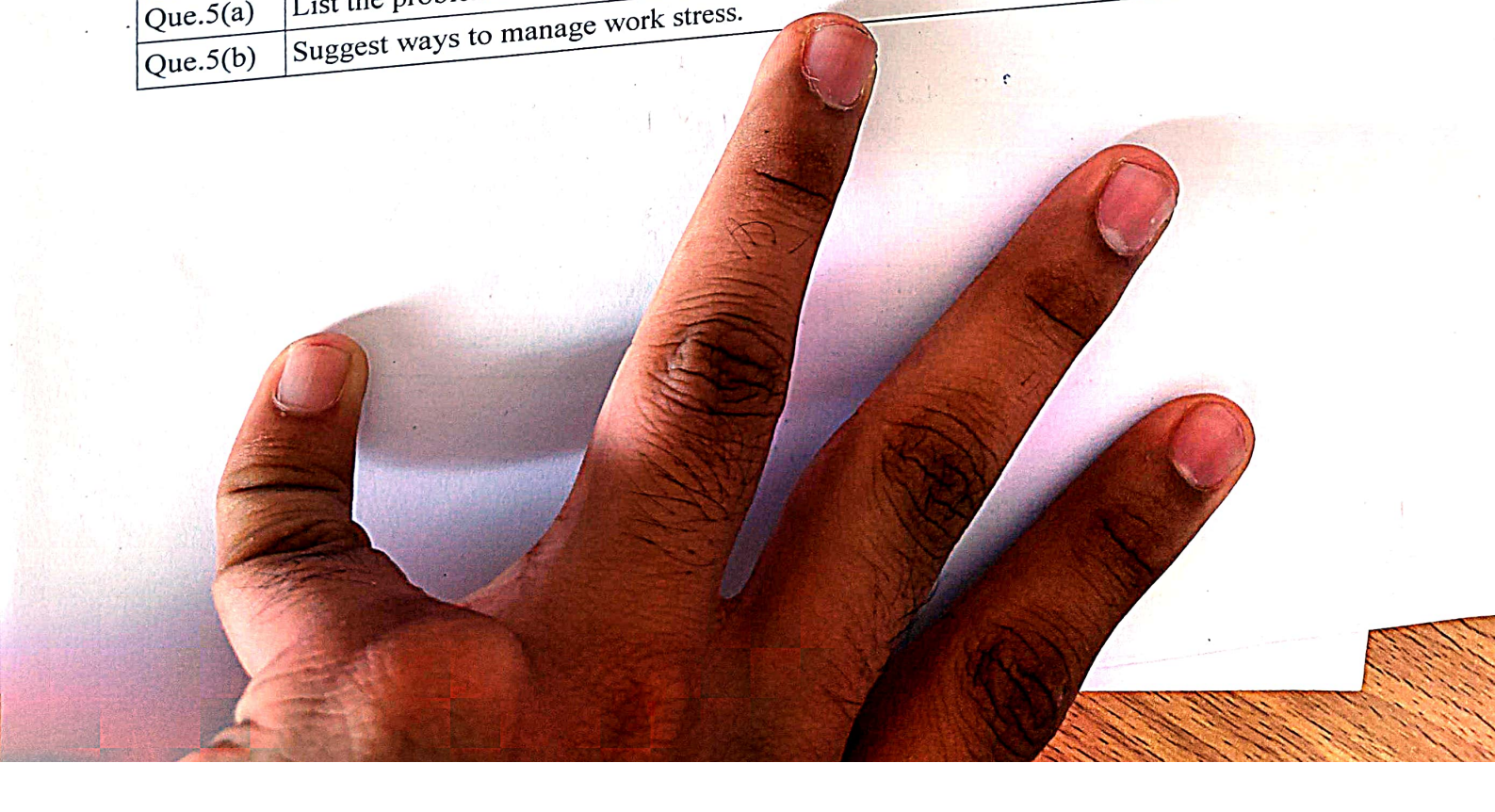
Academic Year: 2024-25

Mid Sem Examination (MSE) - B. Tech

Course: -	**FUNDAMENTALS OF MANAGEMENT**	**Course Code: -**	22CB503T
		Max. Marks: -	20
Semester: -	**V**	**Duration: -**	1 hour
Program: -	**Computer Science & Business Systems**	**Date of Paper**	05/09/24

Instructions to Candidate –
1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) All Questions carry marks as indicated
5) Use of non-programmable calculator is allowed.

Que. No.	Description of Question	Marks	[CO]	[BTL]
Que.1(a)	What is management?	02	1	1
Que.1(b)	Define personality.	02	2	1
Que.1(c)	Discuss the concept of Staffing.	01	1	1
Que.2(a)	Explain the Pre-scientific Era in detail.	04	1	2
Que.2(b)	Enlist eight contributions of management thinker "Taylor" during all eras.	04	1	2
	OR			
Que.3(a)	Describe the Modern Management Era in detail.	04	1	2
Que.3(b)	List the functions of Management.	04	1	1
Que.4(a)	Differentiate between type A and Type B personalities.	04	2	2
Que.4(b)	Describe how cultural diversity is managed.	03	2	2
	OR			
Que.5(a)	List the problems in Decision Making.	04	2	2
Que.5(b)	Suggest ways to manage work stress.	03	2	2



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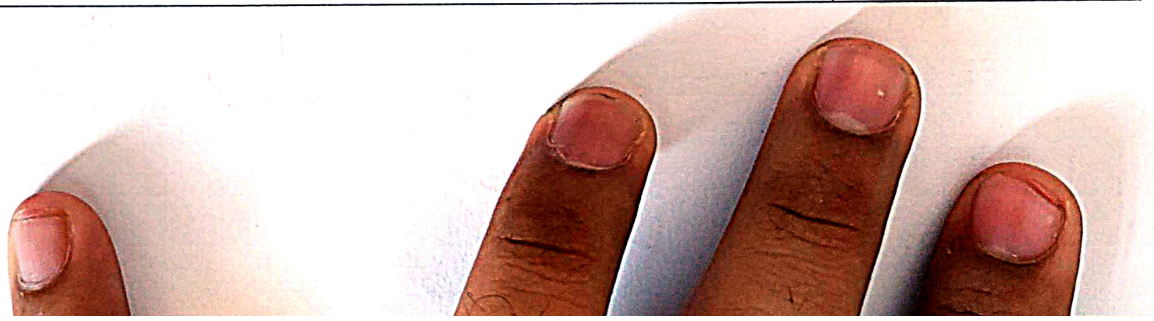
Academic Year: 2024-25

Mid Sem Examination (MSE) - B. Tech

Course: -	**Artificial Intelligence**	**Course Code: -**	22CB603T
		Max. Marks: -	35
Semester: -	**VI**	**Duration: -**	1hr 30mins
Program: -	**Computer Science & Business Systems**	**Date of Paper**	17/03/25

Instructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) All Questions carry marks as indicated

Que. No.	Description of Question	Marks	[CO]	[BTL]
Que.1(a)	Define AI techniques.	02	1	1
Que.1(b)	Demonstrate goal based intelligent agent.	02	2	2
Que.1(c)	Differentiate between uninformed and informed search.	02	3	2
Que.1(d)	What is environment and actuators in AI?	02	1	1
Que.1(e)	Illustrate production system.	02	2	3
Que.2(a)	Discuss intelligent agents and its types.	05	1	2
Que.2(b)	Deduce PEAS representation of AI model with the help of example.	05	1	3
	OR			
Que.3(a)	Summaries different types of production systems used in AI.	05	1	2
Que.3(b)	Discuss AI problem characteristics.	05	1	2
Que.4(a)	Illustrate issues of designing search program.	05	2	2
Que.4(b)	Explain problem state space search with the help of example.	05	2	2
	OR			
Que.5(a)	Demonstrate one possible solution of 8 queen's problem.	05	2	2
Que.5(b)	Explain 8 puzzles problem.	05	2	2
Que.6	Explain BFS techniques with the help of example.	05	3	2
	OR			
Que.7	Summaries goal based, utility and learning agents in detail.	05	1	2



W/24/R

St. Vincent Pallotti College of Engineering & Technology
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B. Tech 5th Semester Computer Science and Business Systems
End Sem Examination

Course:- PE-I-Cloud, Micro services & Application **Course Code:-** 22CB506T(ii)
Duration:- 3 Hours **Max. Marks:-** 70

Instructions to Candidate

1) Question No. 1 is compulsory.
2) Question No. 2 OR 3
3) Question No. 4 OR 5
4) Question No. 6 OR 7
5) Question No. 8 OR 9
6) Question No. 10 OR 11.
7) All Questions carry marks as indicated
8) Use of Non Programmable calculator is allowed.

Q. No.	Description of Question	Marks	[CO]	[BTL]
1(a)	Given a scenario where a company needs scalable storage, which cloud service model would you recommend and why?	02	1	1
1(b)	What are the three main cloud service models?	02	1	1
1(c)	List the main components of Apache Hadoop.	02	2	2
1(d)	What is virtualization?	02	2	1
1(e)	Define cloud security and explain its significance in today's digital landscape.	02	3	2
1(f)	List common risks associated with cloud security	02	3	2
1(g)	Elaborate the meaning of API.	02	4	1
1(h)	Identify the key differences between REST and SOAP APIs.	02	4	4
1(i)	Why DevOps Matters?	02	5	2
1(j)	Compare the roles of developers and operations engineers in a DevOps team.	02	5	4
2(a)	Explain the difference between IaaS, PaaS, and SaaS.	05	1	4
2(b)	What is meant by cloud deployment models? Provide an example.	05	1	2
3(a)	Compare and contrast the advantages and disadvantages of using a public cloud versus a private cloud.	05	1	4
3(b)	Discuss how service-oriented architecture (SOA) can benefit cloud computing applications with the help of suitable diagram.	05	1	2
4(a)	Describe any two algorithms used for load balancing with the help of diagrams.	05	2	2
4(b)	Plot HDFS Structure with all the components and describe role of each component.	05	2	1
5(a)	Represent overall MapReduce word count process for the input given below: *"The quick brown fox jumps over the lazy dog"* *"The fox is quick and the dog is lazy"* *"The fox jumps high Lazy dog do not jump over quick foxes"* *"The dog barked loudly and the fox ran away"*	05	2	3
5(b)	Compare and contrast the benefits and challenges of using virtualization technologies.	05	2	4
6(a)	Describe IAM components. Enlist features of IAM.	05	3	2
6(b)	Identify and describe three key benefits of implementing cloud	05	3	2

	security measures.			
7(a)	Describe the authentication process in cloud environments and explain how it contributes to security.	05	3	2
7(b)	Examine the role of encryption in cloud security and discuss how it helps in protecting data at rest and in transit.	05	3	2
8(a)	Elaborate the application integration process and its significance in Microservices architecture.	05	4	2
8(b)	What are cloud-native applications and why are they important in modern software development?	05	4	1
9(a)	Discuss the design approach for creating microservices and how it differs from traditional development methods.	05	4	2
9(b)	Compare monolithic and distributed architectures.	05	4	4
10(a)	Evaluate how integrating DevOps practices with Docker can enhance the efficiency of cloud application development.	05	5	5
10(b)	Illustrate the core principles of DevOps and express how they improve the software delivery lifecycle.	05	5	2
11(a)	Enumerate three popular tools used in cloud application development and express their primary functionalities.	05	5	2
11(b)	Outline what Docker is and the steps involved in the containerization process.	05	5	2



St. Vincent Pallotti College of Engineering & Technology
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B.Tech 5th Semester Computer Science and Business Systems
End Sem Examination

W/24/R

Course:- Business Strategy **Course Code:-** 22CB504T
Duration:- 1 hour 30 mins **Max. Marks:-** 35

Instructions to Candidate

1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) All Questions carry marks as indicated
6) Use of Non Programmable calculator is allowed.

Q.No.	Description of Question	Marks	[CO]	[BTL]
1(a)	Define Strategic Management.	02	1	1
1(b)	Write short note on 'fit concept' in Strategic Management? Give one example.	02	1	1
1(c)	What do you understand by Core competencies?	02	2	1
1(d)	Define strategy content and explain it with an example.	02	3	2
1(e)	What do you understand by Competitive Advantage?	02	3	1
2(a)	Write down sources of Sustained Competitive Advantage. Explain any 5 in detail.	05	1	2
2(b)	Discuss in detail the objectives of Strategic Management.	04	2	4
3(a)	Explain in detail how Core Competencies can be the roots for Competitive Advantage?	05	1	2
3(b)	Discuss in detail – Schools of thought in Strategic Management.	04	2	4
4(a)	Explain how the five forces of Industry Attractiveness shape the strategies of Industries.	04	3	2
4(b)	Write down in detail the McKinsey's 7 S framework.	04	3	4
5(a)	Discuss how strategic groups are formed. How these groups work together for mutual benefit?	04	3	2
5(b)	Explain the term –Industry Life Cycle. Discuss all the stages in detail.	04	3	4
6(a)	Diversification helps industries to grow – Elaborate	04	4	2
6(b)	Give differentiating points between Expansion and integration.	04	4	4
7(a)	Explain in detail – Strategic Alliances (any two)	04	4	2
7(b)	How does business portfolio analysis helps companies understand how well their products and services are performing and where to invest more money or cut costs	04	4	2

W/24/R

St. Vincent Pallotti College of Engineering & Technology
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B.Tech 5th Semester Computer Science and Business Systems
End Sem Examination

Course:- PE-II Behavioral Economics **Course Code:-** 22CB507T(i)
Duration:- 3 Hours **Max. Marks:-** 70

Instructions to Candidate

1) Question No. 1 is compulsory.
2) Question No. 2 OR 3
3) Question No. 4 OR 5
4) Question No. 6 OR 7
5) Question No. 8 OR 9
6) Question No. 10 OR 11
7) All Questions carry marks as indicated
8) Use of non-programmable calculator is allowed.

Q.No.	Description of Question	Marks	[CO]	[BTL]
1(a)	A person with an infinite capacity for rational decision-making is termed as Homo economicus. True or False?	02	1	01
1(b)	If a person invests in a high-risk stock while also maintaining a savings account for emergencies, that person can be described as: Risk-loving in investments, risk-averse in emergencies. True or False? Explain.	02	1	02
1(c)	The government sent a text message to taxpayers reminding them that "most people pay on time to avoid penalties." This type of message reflects which concept in behavioral economics? Choose one option from below: 1. Risk loving 2. Anchoring 3. Bounded rationality 4. Nudging	02	2	02
1(d)	Bounded rationality suggests that individuals make decisions based on limited information and cognitive limitations, rather than considering all possible options. True or False? Explain.	02	2	02
1(e)	Define rational behavior in economics?	02	3	01
1(f)	Identify at least four biases and beliefs associated with choice theory.	02	3	01
1(g)	Explain the concept of loss aversion within the context of behavioral economics.	02	4	02
1(h)	Which of the following scenarios best illustrates intertemporal choice? 1. Choosing between two brands of the same product. 2. Deciding whether to spend money on a luxury item today or save it for a larger purchase in the future. 3. Selecting a meal from a restaurant menu. 4. Comparing interest rates between two different banks.	02	4	02
1(i)	Define Altruism in terms of behavioral economics.	02	5	01

1(a)	In game theory, the term "Nash equilibrium" refer to? 1. A situation where all players have the same strategy. 2. A scenario in which no player can benefit by changing their strategy while the other players keep theirs unchanged. 3. A situation where all players cooperate for mutual benefit. 4. A condition where players randomly change their strategies.	02	5	01
2(a)	Describe Prospect Theory and demonstrate its principles with examples.	05	1	02
2(b)	Compare and contrast behavioral economics with classical economics.	05	1	04
3(a)	Examine the concept of Money Illusion and provide examples that highlight its importance.	05	1	02
3(b)	Compare Charitable donation, Altruism and behavioral economics.	05	1	04
4(a)	Recognize and elaborate on the fundamental axioms of Choice Theory.	05	2	02
4(b)	Examine the different models of decision making and explain their main attributes.	05	2	02
5(a)	How does the concept of retail therapy relate to choice theory? Explain its connection	05	2	03
5(b)	Describe Public Choice Theory and provide examples to demonstrate its key concepts	05	2	02
6(a)	Explain the concept of Anchoring Effect and give relevant example.	05	3	02
6(b)	Discuss how beliefs, heuristics and biases affect financial trading.	05	3	03
7(a)	Explain heuristics in decision-making and illustrate their use with an example.	05	3	02
7(b)	Compare and contrast common decision-making biases, providing examples for each.	05	3	04
8(a)	Discuss the implications of intertemporal choice on savings behavior and financial planning.	05	4	06
8(b)	Illustrate the concept of loss aversion with an example.	05	4	02
9(a)	Explain hyperbolic discounting with an example.	05	4	02
9(b)	Examine and evaluate different applications of decision-making in uncertain situations.	05	4	04
10(a)	Explain the role of Game Theory in international negotiations.	05	5	02
10(b)	Discuss the principles of Strategic Choice Theory with examples.	05	5	06
11(a)	Explain the concept of Nash Equilibrium in your own words. How does it apply to competitive situations?	05	5	02
11(b)	Explain the disadvantages of using game theory.	05	5	02

S/25/R

W/24/R

St. Vincent Pallotti College of Engineering & Technology
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B.Tech 5th Semester Computer Science and Business Systems
End Sem Examination

Course:- Fundamentals of Management **Course Code:-** 22CB503T
Duration:- 1 Hour 30 Mins **Max. Marks:-** 35

Instructions to Candidate

1) Question No. 1 is compulsory.
2) Question No. 2 OR 3
3) Question No. 4 OR 5
4) Question No. 6 OR 7
5) All Questions carry marks as indicated
6) Use of Non Programmable calculator is allowed.

Q.No.	Description of Question	Marks	[CO]	[BTL]
1(a)	Define Management.	02	1	1
1(b)	Outline the concept of Staffing.	02	1	1
1(c)	State the importance of Reinforcement.	02	2	1
1(d)	List two problems in decision-making.	02	2	2
1(e)	What is a conflict?	02	3	1
2(a)	List Contribution of Management Thinker: Taylor.	05	1	2
2(b)	Compare Directing and Controlling.	04	2	3
3(a)	Explain Modern management Era in detail.	05	1	2
3(b)	Mention techniques of Stress Management.	04	2	3
4(a)	Examine the effectiveness of the Matrix structure in an organization.	04	3	3
4(b)	Describe the Neo-classical approach to organizational design.	04	3	3
5(a)	Illustrate the Simple structure of an organization.	04	3	2
5(b)	Analyze conflict resolution strategies by a manager.	04	3	4
6(a)	Deduce the concept of corporate citizenship.	04	4	3
6(b)	Differentiate between Ethics of Marketing & advertising and Ethics of Finance & Accounting.	04	4	4
7(a)	Elaborate Corporate Governance.	04	4	3
7(b)	Assess the process of developing leaders across the organization.	04	4	5

5/25/R

St. Vincent Pallotti College of Engineering & Technology
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B.Tech VI Semester COMPUTER SCIENCE AND BUSINESS SYSTEMS
End Semester Examination

Course:- Artificial Intelligence **Course Code:-** 22CB603T
Duration:- 3 Hours **Max. Marks:-** 70

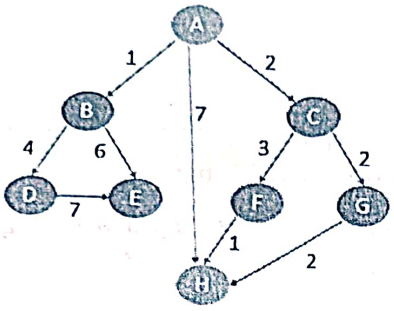
Instructions to Candidate

1) Question No. 1 is compulsory.
2) Solve Question. No. 2 or 3.
3) Solve Question. No. 4 or 5.
4) Solve Question. No. 6 or 7.
5) Solve Question. No. 8 or 9.
6) Solve Question. No. 10 or 11.
7) All Questions carry marks as indicated
8) Use of Non Programmable calculator is allowed.

6TH CSBS

BS-07

Q.No.	Description of Question	Marks	CO	BTL
1(a)	Define goal based intelligent agent.	2	1	1
1(b)	Define environment and sensors in AI.	2	1	1
1(c)	What is problem-solving in the context of Artificial Intelligence?	2	2	1
1(d)	What are the key characteristics of AI problems?	2	2	1
1(e)	Define uninformed search.	2	3	1
1(f)	How will you measure the problem-solving performance?	2	3	1
1(g)	Discuss forward chaining.	2	4	4
1(h)	Elaborate procedural and declarative knowledge.	2	4	2
1(i)	Discuss the key components of an expert system.	2	5	4
1(j)	Write real life application of expert system.	2	5	1
2(a)	Apply AI technique to solve 8-puzzles problem.	5	1	4
2(b)	What is PEAS representation of AI model? Explain with the help of example.	5	1	4
3(a)	Discuss characteristics of intelligent agent and its types.	5	1	4
3(b)	Illustrate four components to define a problem.	5	1	2
4(a)	Discuss production system and its characteristics. Also write about conflict strategies.	5	2	4
4(b)	Define partially commutative and non-partially commutative production system with the help of example.	5	2	4
5(a)	Write all productions rules for solving water jug problem.	5	2	1
5(b)	Discuss design issues of searching problem?	5	2	4

6(a)	Consider following instructions. i) The numbers written on edges represent the distance between the nodes. ii) The numbers written on table represent the heuristic value. iii) Estimate the most cost-effective path to reach from start state A to final state H using A* Algorithm.  Nodes Heuristics A 5 B 3 C 4 D 2 E 6 F 3 G 1 H 0	5	3	6
6(b)	Explain the concept of Bidirectional Search with the help of example.	5	3	2
7(a)	What does AO* Search stand for, and where is it used? Explain the role of the evaluation function in AO* search.	5	3	2
7(b)	Compare DFS and BFS, also write advantages and disadvantages of DFS and BFS.	5	3	4
8(a)	Explain resolution and natural deduction with the help of example.	5	4	2
8(b)	Explain the concept of ISA relationship with an example.	5	4	2
9(a)	Elaborate essential characteristics of Knowledge representation systems? Also write some common schemes of knowledge representation.	5	4	4
9(b)	Construct these sentences in predicate logic. 1. All man drink coffee. 2. Some boys are intelligent. 3. Every student smiles. 4. All Pompeians were Romans. 5. All men are mortal.	5	4	6
10(a)	Discuss expert system. Also write advantages and disadvantages of Expert system.	5	5	4
10(b)	Illustrate Life Cycle of an expert system.	5	5	2
11(a)	Demonstrate architecture of expert system and describe its each component.	5	5	2
11(b)	How do expert systems make decisions, and what role does inference play in them?	5	5	1